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DiS MPHIL

Mechanistic Interpretability of Open Source LLMs

Reproduction of On the Biology of a Large Language Model

A REPORT PRESENTED

BY

IULIIA VITIUGOVA

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Supervision

Dr Miles Cranmer

Dr Alessandro Favero

Departments

Department of Applied Mathematics and Theoretical Physics

Cavendish Laboratory, Department of Physics

Institute of Astronomy

UNIVERSITY OF CAMBRIDGE

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Abstract

The high accuracy of a linear probe is usually read as evidence that a concept is linearly decodable from the hidden states [1], but when the number of examples is far below the hidden dimension a linear classifier can separate the classes under any labelling [2], so the accuracy may reflect the geometry of a small sample rather than the representation [3]. Further, we show that the decoded direction is not always the one the model uses to compute its answer.

In a controlled binary task, the type of nuclear decay, on the Qwen3-4B model with per-layer transcoders, we identify at each layer three directions usually conflated into one, the *reading* direction w_{LDA} (the Fisher discriminant), the *writing* direction $\Delta\mu$ (the difference of class means), and the *use* direction u (the class-averaged gradient of the logit difference). The concept is decoded by w_{LDA} almost perfectly (AUC 0.98 to 0.99), yet intervention along it does not change the answer, while intervention along u or $\Delta\mu$ does. This inertness is controlled by geometry, the reading direction being causal only to the extent that it is non-orthogonal to the use direction, a degree that varies by concept, and with depth the use direction aligns with the unembedding contrast γ while the reading direction stays orthogonal to the output.

Relating the geometry to the transcoder features, the dictionary features write mainly the writing direction rather than the reading axis. At the cluster level they detect physical descriptors of the decay, activating comparably on both classes, so they encode the physics of the process rather than the label, and the class is assembled from them late. The same separation of decoded and used appears at the feature level, the active dictionary spanning the writing direction well above a random null and the use direction partially, but the optimal reading axis no more than the null.

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1 Introduction

Linear representations and linear probing

The linear representation hypothesis states that interpretable features are encoded in the hidden states of neural networks as directions in the activation space [4, 5]. Building on this hypothesis, the standard interpretability method tests for the presence of a concept with a linear classifier, a probe, and high accuracy is interpreted as evidence that the concept is linearly represented [1, 6]. Probe methods have become an established tool, employed to detect part-of-speech, syntactic, and semantic structure [7, 8], and factual associations [9]. Their advantage is that the trained direction serves both as a test of the concept and as a candidate direction in which to intervene, and it is this identification of the test with the mechanism that we question. There is no necessary connection between the *decodability* of a feature and its *causal role*, since a probe can reach high accuracy on a direction that correlates with the label yet is not used by the model when forming the output, measuring statistical separability rather than a computational mechanism [3, 10]. In response, causal methods have been proposed that assess the effect of a direction through intervention rather than through decoding [11–13]. The present work develops this distinction by showing, in a controlled task, that the direction from which the concept is decoded, the reading direction, and the direction to which the output is sensitive, the use direction, are approximately orthogonal, and that both this orthogonality and the causal inertness of reading depend on the concept.

Activation steering and the causal assessment of directions

An alternative to decoding is a causal assessment of a direction through steering. The chosen direction is added to the residual stream at a given layer, and the change in the model's behaviour is measured [14–16]. If the addition systematically shifts the output in the expected way, the direction is considered causally involved. A widely used construction of a steering direction is the difference of the mean activations between classes [17, 18]. This direction, denoted $\Delta\mu$, the writing direction, is the between-class difference written into the residual stream. The third direction is the class-averaged gradient of the margin, the logit difference between the class tokens, with respect to the residual stream. It identifies the direction in which a local change of the residual stream most shifts the model's decision [19]. This direction, denoted u, is the use direction. Because the logits are a linear readout of the residual stream through the unembedding [20], the gradient of the logit difference lies along the unembedding contrast, which makes u a natural candidate for the direction through which the concept reaches the output.

Origin of the study

The project began as a reproduction of Anthropic's circuit tracing methodology [21, 22] on an open model, in which attribution graphs over per-layer transcoder features were constructed and validated by interventions (Appendix B). Two observations from that phase motivated the study of the residual stream directly, at the level of directions. Causal effects were carried by ensembles of features rather than by any single necessary pathway, and many of the features involved activated comparably on both classes, so neither the causal structure nor the class content could be read from individual features.

Transcoder features and hidden organisation

To connect the observed geometry with the model's computations, we use per-layer transcoders as an attribution method, sparse dictionaries that approximate the output of each layer's MLP block as a sum of interpretable feature directions [21, 23]. Whereas sparse autoencoders decompose a layer's activations into monosemantic features [24, 25], transcoders approximate the MLP sublayer itself with a wider, sparsely activating layer, which allows its output to be attributed to individual features [26]. Since each feature contributes to the residual stream through its decoder direction, the

features link the geometry of the directions to the content of the computation. They establish not only that the writing direction lies in the span of the features well above a random null, but what content is written and how it relates to the use direction. Given the polysemanticity of features [27], a single feature is not a reliable unit of analysis, so we group features into clusters by shared causal role, the working unit [22].

A controlled task

The central task is the binary classification of nuclear decay type. Given a description of the process, the model answers α or β. The same concept is presented in three surface forms of increasing context, a direct question, the same question with an answer cue, and the same question with two worked examples and an answer cue. Comparing them separates what is invariant to the surface form from what depends on it. Cross-concept generality is tested on an elementary particle classification, a multiclass task analysed as binary pairs. A full description of the datasets is given in Section 2.3.

Structure

Section 2 describes the methods, covering the model and transcoders, the four geometric objects, the feature metrics and cluster formation, and the intervention protocol and its metrics. **Section 3** presents the main results. These are the causal inertness of reading and its dependence on the concept, the formation of the writing direction by the MLP blocks, the qualitative dissociation of writing and use, the divergence of the directions across depth, the decision zone in which the separation arises, and the physical content of the writing direction. **Section 4** discusses the results in the context of interpretability and linear representations. The auxiliary experiments and additional results form the Supplementary Material, referred to below as Sections S.A through S.F.

2 Methods

2.1 The model and the residual stream

All experiments are conducted on the pre-trained Qwen3-4B base language model [28]. The model is a decoder-only transformer [29] with $L = 36$ layers and hidden dimension $d = 2560$. We use the standard residual stream formulation of the transformer [20], in which the residual stream at position t is updated additively by the contributions of the attention and MLP blocks,

$$h_t^{(\ell)} = h_t^{(\ell-1)} + a_t^{(\ell)} + m_t^{(\ell)}, \tag{1}$$

where $h_t^{(\ell)} \in \mathbb{R}^d$ is the residual stream at the output of layer ℓ for position t, $a_t^{(\ell)}$ is the contribution of the attention block of layer ℓ, and $m_t^{(\ell)}$ is the contribution of its MLP block. The additive structure of (1) permits an exact decomposition of the change of any direction between adjacent layers into attention and MLP contributions (Section 2.6). Throughout, we omit the position index where it is unambiguous, writing $h^{(\ell)}$.

The model's output over the vocabulary, the logits, is formed by applying the unembedding matrix $W_U \in \mathbb{R}^{|V|\times d}$ to the final normalised state.

2.2 Transcoders

To attribute the computation to individual features, we use per-layer transcoders [21, 23]. The layer-ℓ transcoder approximates the mapping from the MLP input to the MLP output of that layer as a sparse combination of feature directions. Given the MLP block input x, the transcoder computes feature activations

$$f^{(\ell)}(x) = \sigma\left(W_{\text{enc}}^{(\ell)} x + b_{\text{enc}}^{(\ell)}\right), \tag{2}$$

and reconstructs the MLP output as

$$\widehat{m}^{(\ell)} = W_{\text{dec}}^{(\ell)} f^{(\ell)}(x) + b_{\text{dec}}^{(\ell)}, \tag{3}$$

where $W_{\text{enc}}^{(\ell)} \in \mathbb{R}^{F\times d}$ and $W_{\text{dec}}^{(\ell)} \in \mathbb{R}^{d\times F}$ are the encoder and decoder matrices, F is the dictionary size, and σ is the activation nonlinearity. We use the pre-trained per-layer transcoders [30] with a dictionary of $F = 163{,}840$ features and a JumpReLU activation [31]. JumpReLU enforces threshold sparsity, a feature being active only if its pre-activation exceeds a trainable threshold.

Each column $W_{\text{dec}}^{(\ell)}[:, j]$ defines the decoder direction of feature j, the direction along which feature j writes its contribution to the residual stream, and the activation $f_j^{(\ell)}$ sets the magnitude of this contribution.

2.3 Datasets and prompt forms

The main corpus, the α/β decay corpus, consists of descriptions of nuclear processes labelled α or β, class-balanced with 269 examples per class for a total of 538.

The same concept is presented in three prompt forms.

1. The **raw** form is a direct question about the decay type, with the class label as the target answer.

2. The **raw with suffix** form is a direct question with an added answer cue, a suffix that fixes the response format by requesting the answer as α or β.

3. The **scaffold** form prepends two labelled examples, each with the answer cue, before the target question [32]. These in-context examples fix the output format, indicating the expected answer form without otherwise changing the task (Section 3.2).

Because all three express the same α/β concept, agreement of the measured geometry across forms reflects invariance to the surface form rather than a coincidence of distinct concepts.

A second task tests cross-concept generality. The **elementary particle** task is a multiclass problem over electron, neutron, photon, and proton, with a corpus of 320 examples balanced across the four particles. From this corpus we form the six binary pairs, all pairs of the four classes, each balanced with 80 examples per class. The binary pairs test the invariance of the writing mechanism to the choice of pair within a single concept (Section 3.2).

2.4 Measurement position

All geometric objects are measured in the residual stream at the answer position, where the model generates the class token. The prompt forms (Section 2.3) fix the output format, making this position unambiguous and comparable across examples.

2.5 Four geometric objects

The analysis concerns four directions in the residual stream, three defined at each layer ℓ and the layer independent unembedding direction γ (Figure 1).

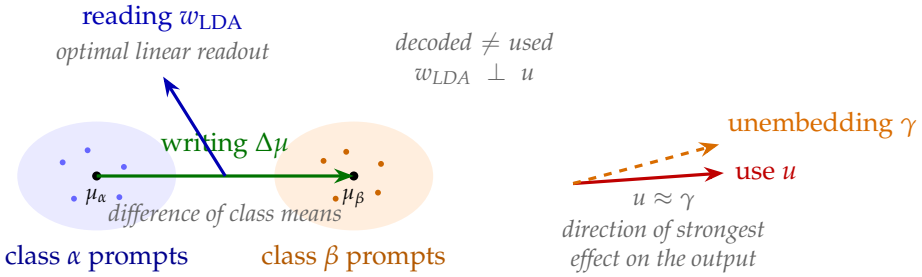


Figure 1: The four geometric directions.

2.5.1 Writing direction $\Delta\mu$

The writing direction is defined as the difference of the mean residual stream between classes at layer ℓ,

$$\Delta\mu^{(\ell)} = \mu_{c_1}^{(\ell)} - \mu_{c_0}^{(\ell)}, \qquad \mu_c^{(\ell)} = \frac{1}{|D_c|} \sum_{x \in D_c} h^{(\ell)}(x), \tag{4}$$

where D_c is the set of examples of class c and $\mu_c^{(\ell)}$ is the class mean of the residual stream. Thus $\Delta\mu^{(\ell)}$ is the between-class difference written into the stream, the direction along which the mean representations of the two classes diverge [17, 18].

The difference of means construction subtracts the part of the representation common to both classes. The remaining direction $\Delta\mu$ retains a substantial component along the mean residual stream and is large in norm, so we compare directions only after unit normalisation and track the norm of $\Delta\mu$ separately (Section 3.4). The normalised writing direction is $\widehat{\Delta\mu}^{(\ell)} = \Delta\mu^{(\ell)}/\|\Delta\mu^{(\ell)}\|$.

2.5.2 Reading direction $w_{\mathbf{LDA}}$

The reading direction is the Fisher linear discriminant [33, 34]. Fisher LDA seeks the direction w that maximises the ratio of between-class to within-class variance,

$$J(w) = \frac{\left(w^\top(\mu_{c_1} - \mu_{c_0})\right)^2}{w^\top S_W\, w}, \tag{5}$$

where $S_W^{(\ell)}$ is the pooled within-class covariance,

$$S_W^{(\ell)} = \frac{1}{|D| - 2} \sum_{c \in \{c_0, c_1\}} \sum_{x \in D_c} \left(h^{(\ell)}(x) - \mu_c^{(\ell)}\right)\left(h^{(\ell)}(x) - \mu_c^{(\ell)}\right)^\top. \tag{6}$$

The maximiser of (5) is $w \propto S_W^{-1}\Delta\mu$, which we compute with the regularised covariance Σ_s,

$$w_{\mathrm{LDA}}^{(\ell)} \propto \left(\Sigma_s^{(\ell)}\right)^{-1}\Delta\mu^{(\ell)}, \tag{7}$$

where $\Sigma_s^{(\ell)} = (1 - s)S_W^{(\ell)} + s\ \mathrm{diag}\big(S_W^{(\ell)}\big)$. With $|D| = 538$ examples in $d = 2560$ dimensions, the within-class covariance is rank-deficient and not invertible. We therefore regularise it by shrinkage toward its diagonal [35], which shrinks the within-class correlations while keeping the per-feature variances, at a fixed shrinkage intensity $s = 0.1$ [36]. This intensity is fixed a priori for all analyses. The whitening attenuation that turns the probe away from the use direction weakens monotonically as the intensity is raised, so a less regularised probe would be no less orthogonal, and the reported orthogonality is not produced by the regularisation (Appendix A).

By (7), w_{LDA} is $\Delta\mu$ whitened by the inverse within-class covariance, the direction of maximal linear class separability. We call it the reading direction because it is the optimal linear readout of the class. The normalised reading direction is

$$\widehat{w}_{\mathrm{LDA}}^{(\ell)} = \frac{\left(\Sigma_s^{(\ell)}\right)^{-1}\Delta\mu^{(\ell)}}{\left\|\left(\Sigma_s^{(\ell)}\right)^{-1}\Delta\mu^{(\ell)}\right\|}.$$

2.5.3 Use direction u

The use direction is obtained by differentiating the class margin with respect to the residual stream. We define the margin as the difference between the logits of the two class tokens,

$$\mathrm{margin}(x) = \mathrm{logit}_{c_1}(x) - \mathrm{logit}_{c_0}(x), \tag{8}$$

where $\mathrm{logit}_c(x)$ is the logit of the class c token at the answer position. The margin gradient with respect to the residual stream at layer ℓ points in the direction in which a local change of the residual stream most strongly increases the preference for c_1 over c_0 [19]. The class-averaged use direction is

$$u^{(\ell)} = \frac{1}{2}\left(\frac{1}{|D_{c_0}|}\sum_{x \in D_{c_0}} \nabla_{h^{(\ell)}}\mathrm{margin}(x) + \frac{1}{|D_{c_1}|}\sum_{x \in D_{c_1}} \nabla_{h^{(\ell)}}\mathrm{margin}(x)\right). \tag{9}$$

The balanced average over the two classes in (9) gives the direction to which the model's decision is on average most sensitive, regardless of the correct class. The normalised use direction is

$$\widehat{u}^{(\ell)} = \frac{\overline{\nabla_{h^{(\ell)}}\,\mathrm{margin}}}{\left\|\overline{\nabla_{h^{(\ell)}}\,\mathrm{margin}}\right\|},$$

where the overbar denotes the class average, as in (9).

2.5.4 Unembedding direction γ

The unembedding direction is the contrast in the unembedding space between the class tokens,

$$\gamma = W_U[c_1, :] - W_U[c_0, :], \tag{10}$$

where $W_U[c, :]$ is the row of the unembedding matrix corresponding to the class c token [5]. Equivalently, γ is a direction in the residual stream space, and the projection of the final normalised state onto γ is the logit difference (8). We use γ as a reference for interpretation. The normalised unembedding direction is $\widehat{\gamma} = \gamma / \|\gamma\|$.

2.6 Decomposition of the change of a direction

The change of a direction between adjacent layers is decomposed in two complementary ways. Geometrically, an increment $\delta v = v^{(\ell)} - v^{(\ell-1)}$ splits into a length component along the current direction and a rotation component orthogonal to it,

$$\mathrm{len}(\delta v) = \delta v^{\top} \widehat{v}^{(\ell-1)}, \qquad \mathrm{rot}(\delta v) = \left\| \delta v - \mathrm{len}(\delta v)\, \widehat{v}^{(\ell-1)} \right\|, \tag{11}$$

separating changes that rescale a representation from those that reorient it. The additive structure of the residual stream (1) splits the change of the writing direction into an attention and an MLP contribution. Since $\Delta\mu$ is a linear function of the class means, which inherit that additivity,

$$\Delta\mu^{(\ell)} - \Delta\mu^{(\ell-1)} = \underbrace{\left(\bar{a}^{(\ell)}_{c_1} - \bar{a}^{(\ell)}_{c_0}\right)}_{\text{attention}} + \underbrace{\left(\bar{m}^{(\ell)}_{c_1} - \bar{m}^{(\ell)}_{c_0}\right)}_{\text{MLP}}, \tag{12}$$

with $\bar{a}^{(\ell)}_c$ and $\bar{m}^{(\ell)}_c$ the class-averaged attention and MLP contributions of layer ℓ. The identity holds to machine precision and uses the true MLP output captured on the forward pass, not its transcoder reconstruction, which is unreliable on the small class difference. Attributing the rotation of (11) to each block through (12) gives the MLP share of the rotation,

$$\mathrm{MLP\%_{rot}} = \frac{\left\langle |\mathrm{rot_{MLP}}| \right\rangle}{\left\langle |\mathrm{rot_{MLP}}| + |\mathrm{rot_{attn}}| \right\rangle}, \tag{13}$$

where $\mathrm{rot}_B(\delta v) = \langle \delta v_B, \widehat{r} \rangle$ is the contribution of block $B \in \{\mathrm{MLP}, \mathrm{attn}\}$ projected onto the unit rotation direction $\widehat{r} = \left(\delta v - \mathrm{len}(\delta v)\, \widehat{v}^{(\ell-1)}\right) / \mathrm{rot}(\delta v)$, with δv_{MLP} and δv_{attn} the two terms of (12). The signed projections are additive, $\mathrm{rot_{MLP}} + \mathrm{rot_{attn}} = \mathrm{rot}(\delta v)$, so the share in (13) is taken over the summed absolute block projections, which equal the total rotation when the two blocks turn the direction in the same sense. The share is averaged over the layers where the rotation is substantial (fraction above 0.3), and the length share likewise. The decomposition (12) is mechanically exact only for the writing direction, which is linear in the residual stream. The use and reading directions are nonlinear, so for them the split is only a geometric alignment, and we claim MLP formation only for $\Delta\mu$ (Section 3.2).

2.7 Direction interventions

We assess the causal role of a direction by activation intervention [14–16]. At layer ℓ, at the answer position, we add it to the residual stream,

$$h^{(\ell)} \leftarrow h^{(\ell)} + s\, c\, \sigma^{(\ell)}\, \widehat{v}, \qquad \widehat{v} \in \left\{ \widehat{\Delta\mu}^{(\ell)},\ \widehat{u}^{(\ell)},\ \widehat{w}^{(\ell)}_{\mathrm{LDA}} \right\}, \tag{14}$$

where $\widehat{v}$ is a unit direction, $s \in \{-1, +1\}$ sets the sign toward the target class, $c \in \{1, 4, 8, 16, 32\}$ is the strength, and $\sigma^{(\ell)}$ is the standard deviation of the activation projections at layer ℓ [16], computed per layer so that a given c is comparable at every depth, and shared across directions so the comparison reflects the direction rather than the perturbation size. The intervention is applied per direction at every layer $\ell \in \{1, \ldots, 36\}$, giving a causal profile of each direction across layers.

2.8 Feature attributions

We measure the contribution of each feature to the answer by gradient times activation [21, 26]. For prompt p and feature k at layer ℓ,

$$\tau_k^{(\ell)}(p) = a_k^{(\ell)}(p)\,\langle W_{\text{dec}}^{(\ell)}[k],\, g^{(\ell)}(p)\rangle, \qquad g^{(\ell)}(p) = \frac{\partial\,\text{margin}(p)}{\partial\, m^{(\ell)}}, \tag{15}$$

where $a_k^{(\ell)}(p)$ is the activation of feature k, $W_{\text{dec}}^{(\ell)}[k]$ is its decoder direction, $m^{(\ell)}$ is the MLP output, and $g^{(\ell)}(p)$ is the gradient of the margin with respect to the MLP output. The inner product $\langle W_{\text{dec}}^{(\ell)}[k], g^{(\ell)}(p)\rangle$ is the effect on the margin of the content the feature writes, so attribution combines how strongly a feature activates with the first-order effect of its written direction on the margin.

Attribution separates presence from use. A feature can activate strongly yet have near zero attribution if its decoder direction is orthogonal to the gradient, present in the activations but not moving the margin to first order.

We also use attribution to select the features for the rest of the analysis. The features with the largest attribution form an attribution graph, and these are the features we ablate (Section 2.9) and group into clusters (Section 2.10).

This attribution graph is the basis of the reproduction. Graphs over transcoder features were built on four corpora and validated by feature ablation and activation patching, confirming that the methodology of [21, 22] transfers to the open model, producing compact chains whose nodes carry readable object features. Appendix B reports the reproduction in full.

2.9 Feature ablation

The causal role of a feature or of a cluster of features is tested by ablation, removing it and measuring the change in the output [26]. We subtract the feature's contribution $a_f^{(\ell)}(p)\, W_{\text{dec}}^{(\ell)}[:, f]$ from the true MLP output, so the intervention does not pass through the transcoder reconstruction, and the forward pass continues through the remaining layers. The effect on prompt p is the signed change in the margin,

$$e(f, p) = \text{margin}_{\text{ablated}}(p) - \text{margin}_{\text{baseline}}(p), \tag{16}$$

and we summarise a feature by the mean absolute effect over prompts and by the relative effect $|e|/|\text{margin}_{\text{baseline}}|$. These per-feature effects are the input to the feature clustering (Section 2.10).

2.10 Cluster formation

Because a single feature is polysemantic [27], we group features by their joint causal importance. For each feature we take the ten prompts on which its ablation has the largest absolute effect (Section 2.9), and define the co-importance of two features as the Jaccard overlap of these prompt sets,

$$\text{coimp}(i, j) = \frac{|T_i \cap T_j|}{|T_i \cup T_j|}, \tag{17}$$

where T_i is the set of the ten prompts with the largest ablation effect for feature i. We convert co-importance into a distance $D_{ij} = 1 - \text{coimp}(i, j)$ and cluster the features by hierarchical agglomerative clustering with average linkage [37]. The distance between two clusters A and B is the mean pairwise distance between their features,

$$d(A, B) = \frac{1}{|A|\,|B|} \sum_{i \in A} \sum_{j \in B} D_{ij}, \tag{18}$$

and at each step the two clusters with the smallest $d(A, B)$ are merged, starting from each feature as its own cluster. We cut the resulting hierarchy at $K = 12$ clusters, the finest resolution whose clusters

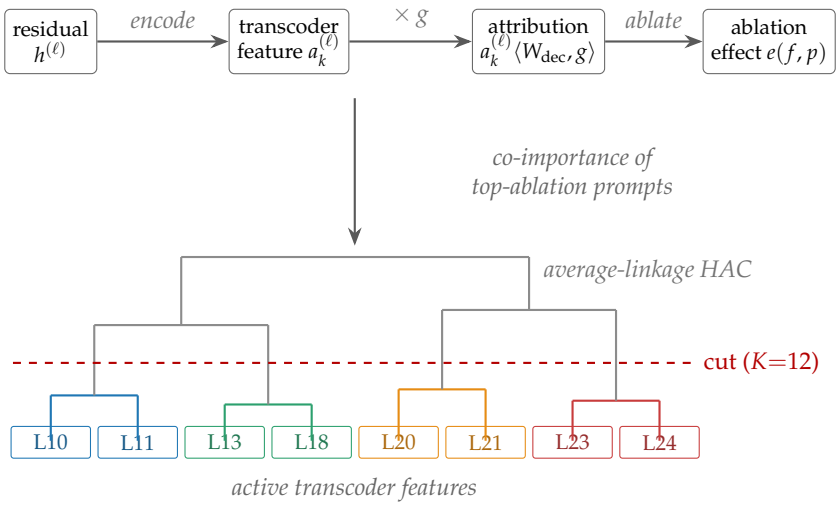


Figure 2: Path of a transcoder feature into a cluster.

stay as stable under paraphrasing of the input as a random partition (Supplementary Material, Section S.D), to obtain the clusters used as the working unit in Section 3.6 (Figure 2).

3 Results

3.1 Geometry of directions

3.1.1 Decodability

The reading direction w_{LDA} (7) decodes the decay type almost perfectly, with a held-out AUC reaching 0.99 from the middle layers onward (Table 1). The writing direction $\Delta\mu$ is also decodable, with an AUC up to 0.96, so high decodability is not specific to the reading direction [1, 6]. The use direction u, by contrast, stays near chance in the middle layers and becomes decodable only near the output, a contrast taken up in Section 3.1.2.

layer	w_{LDA}	$\Delta\mu$	u
4	0.938	0.784	0.638
12	0.967	0.870	0.594
18	0.994	0.955	0.506
22	0.999	0.965	0.516
24	0.998	0.962	0.700
28	0.998	0.943	0.623
32	0.998	0.915	0.718
35	0.993	0.917	0.877

Table 1: Held-out AUC per layer, α/β.

3.1.2 Causal inertness

Despite high decodability, intervention along w_{LDA} leaves the answer almost unchanged on the α/β task. We measure the causal effect by the margin flip, the fraction of examples whose class margin reverses sign,

$$\text{margin flip} = \mathbb{E}_x\big[\ \text{sign margin}'(x) \neq \text{sign margin}(x)\ \big], \tag{19}$$

where $\text{margin}(x)$ is the quantity of Equation 8 and $\text{margin}'(x)$ is its value after the intervention. We use the class margin rather than the top token, which is format-dependent and undercounts flips that produce coherent text rather than a bare class token.

At strength $c = 16$ the causal effect of w_{LDA} depends on the concept and tracks its geometry (Table 2). On α/β the reading direction is approximately orthogonal to the use direction ($\cos = 0.02$) and causally inert, with a margin flip of 0.07 against a random control of 0.01 (a random unit direction at the same strength). On the particle concept the reading direction is non-orthogonal to use ($\cos = 0.15$) and causal, with a margin flip of 1.00 against a control of 0.23. The writing direction, non-orthogonal to u on both concepts, and the use direction itself flip the margin in both cases. The reading direction is therefore inert where it is orthogonal to the use direction and causal where it aligns with it, so its effect is mediated by its component along u. This concept dependence is not an artefact of the strong intervention. At $c = 4$, where the random control is still near zero, the reading direction flips 0.50 on the particles against 0.04 on α/β, and it stays below 0.08 on α/β at every strength, so the effect of reading tracks its alignment with u rather than saturation.

The dissociation is symmetric. The reading direction decodes the class almost perfectly yet leaves the answer unchanged, while the projection onto the use direction decodes barely above chance in the decision zone (AUC 0.516 at layer 22, Table 1) at the very layers where intervention along it reverses the margin on 0.88 of the examples (Table 2). One direction carries the class information without causal effect, the other the causal effect without readable class information.

concept	$\cos(w_{\mathrm{LDA}}, u)$	$\cos(\Delta\mu, u)$	w_{LDA}	$\Delta\mu$	u	random
α/β	0.02	0.34	0.07	0.88	0.88	0.01
particles	0.15	0.26	1.00	1.00	1.00	0.23

Table 2: Geometry and causal effect of the three directions.

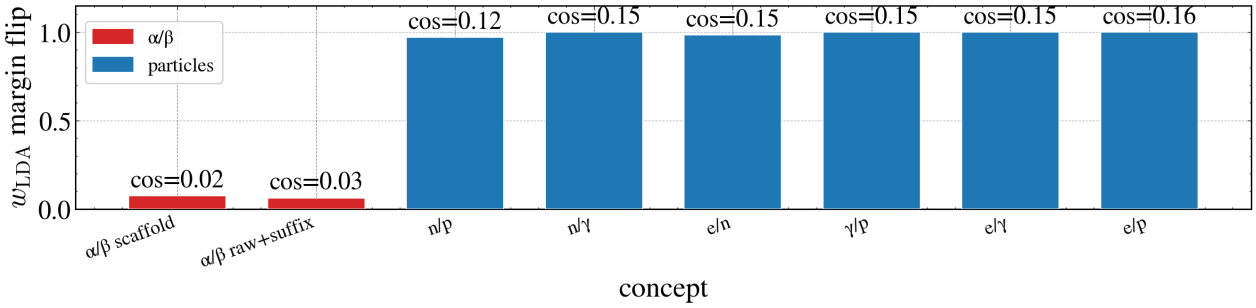


Figure 3: Reading direction margin flip against alignment with use, across concepts.

3.1.3 Orthogonality and output alignment

The reading and use directions are approximately orthogonal, with a cosine averaged over layers of $\cos(w_{\mathrm{LDA}}, u) \approx 0.02$. This orthogonality is not an empirical accident but a consequence of an exact attenuation identity, derived in Appendix A, by which the whitening turns the optimal probe away from the high-dispersion use direction. Because a small cosine is expected in high dimension, this orthogonality is assessed against a covariance-matched null and an attenuation bound rather than the isotropic baseline, against which the measured cosine sits at the 18th percentile and its reliability-corrected upper bound never exceeds 0.074 (Appendix A).

The alignment with γ confirms this (Table 3). With depth the use direction converges to it, reaching $\cos(u, \gamma) = 0.99$ at the final layer. The writing direction aligns partially, its cosine with γ peaking at 0.42 and sitting at 0.33 at the final layer, and the reading direction stays orthogonal to γ, with $\cos(w_{\mathrm{LDA}}, \gamma) \approx 0.01$. Because the projection onto γ is the logit difference (8), the alignment of u with γ and the orthogonality of w_{LDA} to γ are independent evidence that the concept enters the prediction through u, not through w_{LDA}.

The geometry across layers matches the causal effect (Table 3). The reading direction stays orthogonal to both u and γ at every layer and its margin flip never exceeds 0.07. The use direction aligns with γ first and is causal earliest, with a margin flip of 0.41 at layer 20. The writing direction aligns with use around layers 21 to 22 and becomes causal there, with a margin flip of 0.85 at layer 22. These onsets fall within a narrow band of layers, the decision zone, the band of layers 19 to 24 where decodability along u, the causal effect, and the alignment of writing with use first exceed their nulls, centred on L21.

Because u is the gradient of the margin, a first-order step along it raises the margin by construction, so the identity is not itself a finding. Yet three of the results above do not follow from it. The reversals are measured at a finite step of 16σ, realised through the full nonlinear network rather than the first-order term (Table 2). The alignment of u with γ forms within the narrow decision zone at L21 rather than uniformly with depth (Table 3). The equality $u = \gamma$ at the readout is definitional, but the depth at which the alignment forms is not. The reading direction w_{LDA} is defined from the class covariance (7), is independent of u, decodes the class at AUC 0.99 (Table 1), and yet its margin flip does not exceed 0.07 (Table 2), which no gradient identity forces. The contrast between the direction that reads the class and the direction that produces it is the finding, and the definition of u does not supply it.

L	$\cos(w,u)$	$\cos(\Delta\mu,u)$	$\cos(u,\gamma)$	$\cos(w,\gamma)$	$\cos(\Delta\mu,\gamma)$	w	$\Delta\mu$	u
16	0.03	0.06	0.04	0.03	0.00	0.00	0.00	0.05
20	0.02	0.09	0.05	0.03	0.01	0.03	0.12	0.41
22	0.03	0.29	0.14	0.01	0.02	0.06	0.85	0.88
24	0.00	0.40	0.27	0.01	0.14	0.03	0.88	0.88
28	0.03	0.39	0.47	0.02	0.15	0.05	0.88	0.88
32	0.01	0.42	0.64	0.01	0.42	0.03	0.88	0.88
35	0.03	0.34	0.99	0.03	0.33	0.01	0.88	0.88

Table 3: Cosines and margin flip per layer ($c = 16$), α/β.

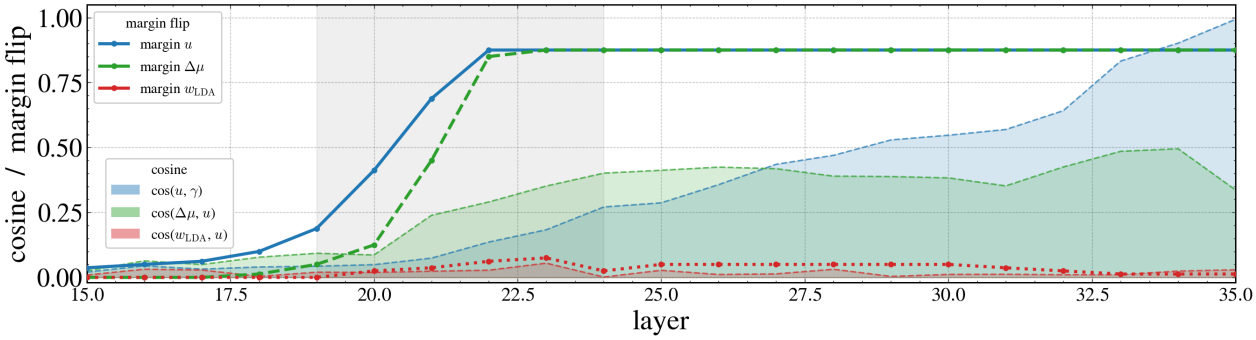


Figure 4: $\cos(\cdot,\gamma)$ and margin flip across layers, α/β.

3.2 MLP blocks and writing direction

Decomposing the rotation of the writing direction into blocks on the true MLP output (Section 2.6) shows that the MLP blocks carry most of it, and the result is invariant in two independent ways (Table 4). Across the three framings of the α/β task the MLP share of the rotation stays above one half, from 56% on the raw form with suffix to 70% on the raw form, so the mechanism tracks the concept rather than the surface form. Across the six particle pairs it lies in the narrow band 82 to 86%, so it is not tied to one pair either. The MLP share of the rotation across the pairs is far more tightly grouped than its share of the length (42 to 62%), so the rotation is the purer MLP effect while attention contributes more to the length.

The MLP contribution shifts with depth (Figure 5). In the early layers the MLP carries more of the length of the writing direction than of its rotation, and in the late layers more of the rotation, a crossover present on every concept. The late layers, where the rotation of the writing direction is most MLP driven, are also where the writing direction is formed, 91% of it in layers 27 to 35 (Section 3.5). The block decomposition and the feature capture of Section 3.5 are therefore two views of the same late formation, the MLP blocks that form the writing direction and the transcoder features that capture it above the random null.

The gap between the two bands is left open, the concepts differing in corpus size and prompt structure.

concept	rotation	length
α/β raw	70	66
α/β scaffold	60	40
α/β raw with suffix	56	52
electron/photon	86	62
electron/neutron	85	53
electron/proton	85	55
neutron/photon	84	53
photon/proton	85	52
neutron/proton	82	42

Table 4: MLP share of the rotation and length of $\Delta\mu$, α/β.

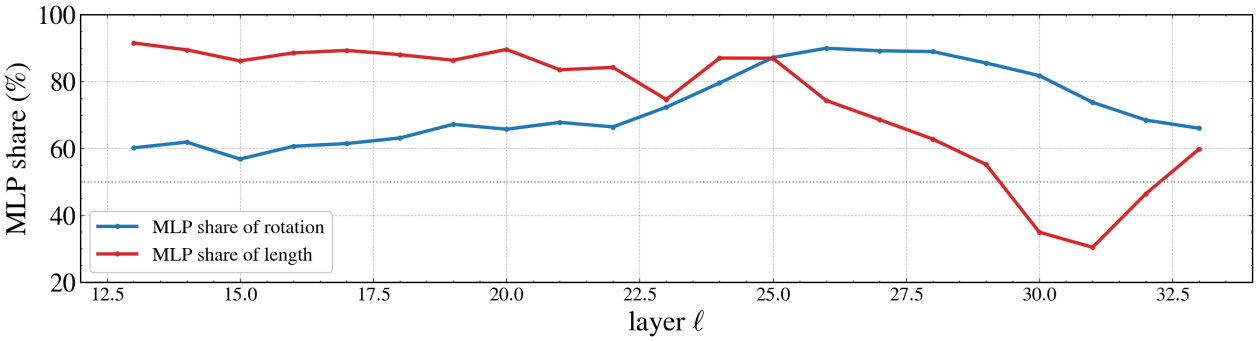


Figure 5: MLP share of the rotation and length of $\Delta\mu$ across layers, α/β.

3.3 Dissociation under intervention

The three directions differ in their effect on the generation under intervention. The writing direction $\Delta\mu$ and the use direction u flip the class margin and are causal (Section 3.1.2), while the reading direction w_{LDA} is inert at moderate strength and moves the margin only at large strength, through its small component along the use direction (Section 3.1.2). Their effect on the generation still differs. We steer at the answer position of a late layer (L24) on a particle prompt and read the continuation as the strength c grows (Figure 6).

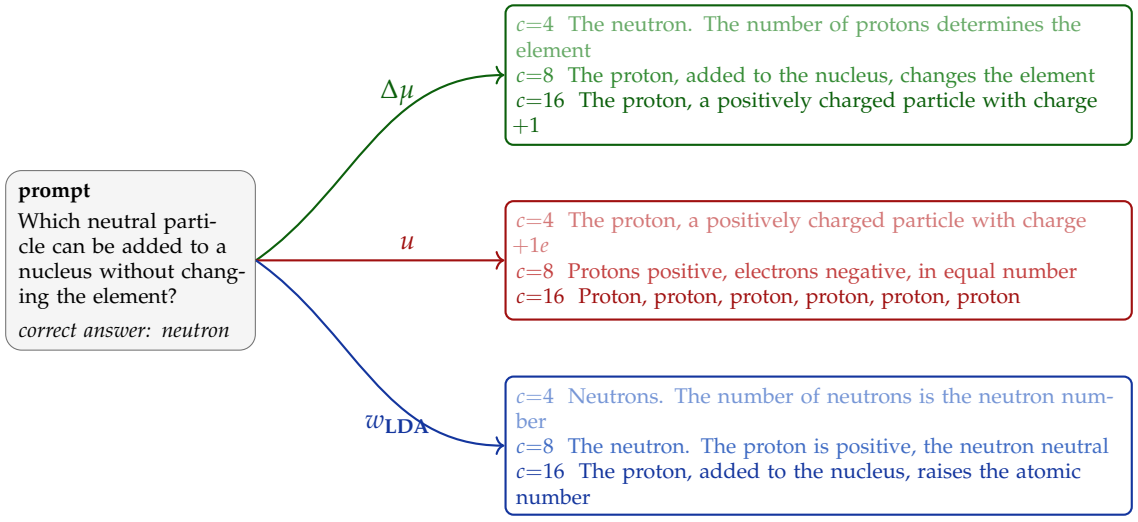


Figure 6: Steered generations, particle (neutron/proton).

The use direction forces the class token and, as the strength grows, degenerates the continuation

into a repetition of it, because it aligns with the unembedding contrast γ toward the output (Section 3.1.3), the direction that sets the class token logits. The writing and reading directions are orthogonal to γ and write content into the residual stream rather than raising a single token logit, so their continuations stay coherent rather than collapsing into a repeated token. Of the three the reading direction changes the answer last, holding the correct class to a higher strength than $\Delta\mu$ and u before it too flips at the largest strength, its margin on the particles turning over as well (Table 2). A top-token metric would miss this content, so the comparison uses the class margin and the generation.

Each steering direction is a fixed vector, the between-class contrast estimated on held-out prompts of the pair, and adding it moves the answer to the paired particle rather than to an arbitrary class, even though the prompt does not mention it. This specificity is consistent with the categorical geometry of concepts [38]. In that account the classes of a categorical concept form a simplex in the representation space, and the between-class contrast is the edge between two vertices, so a push along $\Delta\mu$ moves the state along that edge toward the neighbouring vertex and not toward the other classes.

3.4 Comparison of direction norms

Sections 3.1 and 3.3 separate the writing and use directions by their causal role. This section adds a geometric description. Relative to the norm of the residual stream, the two directions behave oppositely with depth, which gives a mechanistic reading of the content versus token dissociation of Section 3.3. We measure the ratio of each direction's norm to the residual stream norm by layer,

$$r_v^{(\ell)} = \frac{\|v^{(\ell)}\|}{\|h^{(\ell)}\|}, \qquad v \in \{\Delta\mu, u\},$$

comparing the early layers ($\ell < 22$) with the later layers ($\ell \geq 22$), for several concepts (Table 5). The writing direction $\Delta\mu$ grows relative to the stream toward the output, from 0.05 to 0.35 on the scaffold and from 0.16 to 0.47 on the particles, while the use direction u shrinks. The growth of the writing direction is framing-dependent and is weaker on the raw form with suffix (0.024 to 0.14) than on the scaffold, but the shrinking of the use direction is present on every concept tested. The gap between the two norms reaches between two and three orders of magnitude toward the output, where $\|\Delta\mu\|$ is up to about 900 times $\|u\|$.

concept	$r_{\Delta\mu}$ early→zone	r_u early→zone	$\|\Delta\mu\|/\|u\|$ (L35)
α/β scaffold	$0.05 \to 0.35$	$0.023 \to 0.003$	726
α/β raw with suffix	$0.024 \to 0.14$	$0.075 \to 0.004$	234
particles (e/γ)	$0.16 \to 0.47$	$0.013 \to 0.003$	903

Table 5: Norm of $\Delta\mu$ and u relative to the residual stream, and their ratio at L35.

This gives a mechanistic reading of the dissociation. The between-class difference is written into the stream as content. Toward the output it grows in norm and takes up an increasing share of the stream. The use direction requires not a large norm but a precise orientation, so it shrinks relative to the stream while remaining the direction to which the output is sensitive (Section 3.1.3). The writing direction grows into the stream, the use direction shrinks within it.

This geometry separates the two directions in a way their rotation does not. Both rotate by a similar angle across depth, the common background rotation of the task subspace, so it is not the rotation that distinguishes them but their relation to the stream norm.

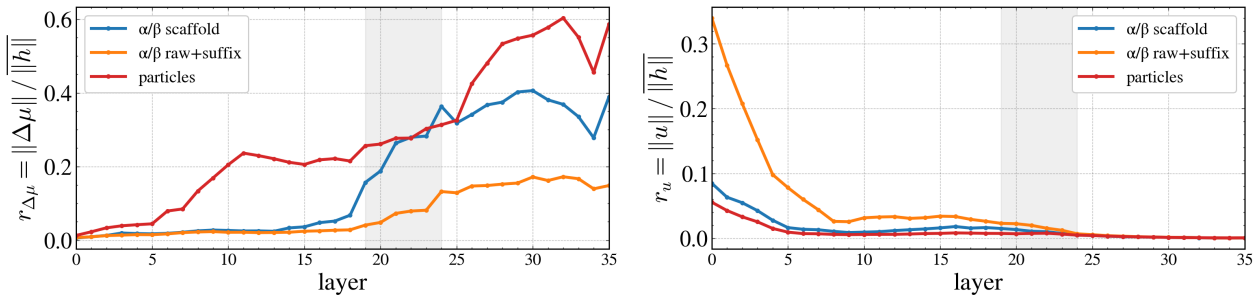


Figure 7: Norm of $\Delta\mu$ and u relative to the residual stream across layers.

3.5 Feature span projection on the directions

The reading, writing, and use directions differ in whether the transcoder features can represent them. We restrict to the active subdictionary A, the features active on at least 5% of prompts, and measure how much of each unit direction lies in the span of their decoder directions,

$$\text{capture}(v) = \frac{\|P_{\text{span}(D_A)}\,\hat{v}\|^2}{\|\hat{v}\|^2}, \qquad D_A = \{\, W_{\text{dec}}[f] : f \in A^{(\ell)} \,\},$$

where P is the orthogonal projection onto the column space of D_A. Capture is 1 when $\hat{v}$ lies entirely in the active span and 0 when it is orthogonal to it. The random null is the same quantity for a random unit vector, which in expectation equals $|A|/d$, the fraction of any direction that an $|A|$-dimensional subspace captures. Here it is about 0.015 to 0.035, matching $|A| \approx 40$ to 90 out of $d = 2560$ (Table 6).

layers	$\lvert A\rvert$	$\Delta\mu$	u	w_{LDA}	null	$\Delta\mu$/null	u/null	w/null
8–15	38	0.054	0.021	0.019	0.015	3.6	1.4	1.3
20–26	87	0.207	0.077	0.045	0.035	5.9	2.2	1.3
30–35	43	0.242	0.044	0.027	0.016	15.1	2.7	1.7

Table 6: Active feature span capture by layer band.

The writing direction is captured well above the null, by a factor of about 6 in the middle band (capture 0.207 against a null of 0.035) and 15 in the late layers. The use direction is captured partially, by about 2.2. The reading direction remains within a factor of 1.3 to 1.7 of the null across bands, far below the factors 6 to 15 of the writing direction. What the features write lies within the reach of the dictionary, while the optimal reading direction does not. This is the feature-level counterpart of the geometry of Section 3.1.3. The whitening rotates the reading direction out of the subspace in which the MLP writes, consistent with the capture at the null (Appendix A).

Capture is computed with free coefficients, so it establishes that a direction is expressible within the span of the active decoder directions, not that the features write it with their realised activations.

The absolute capture is modest, around 0.2 to 0.3 for the writing direction, because the active subdictionary is small, 10 to 135 features per layer, and the between-class difference partly lies outside the span of the features. The result is the contrast with the null.

The population of writing features is not static across depth. Each layer has its own dictionary, and the number of features that write the between-class difference expands in the middle band (L20 to L26), from about 8 in the early layers to 56 in layers 20 to 26, where the writing is also most distributed, the five strongest features accounting for 70% of the written difference against about 95% in the early and late layers. Most of the contribution to $\Delta\mu$ occurs in the late layers, 91% of it in layers 27 to 35, the same layers where the MLP blocks dominate the rotation of the writing direction (Section 3.2). The writing is thus a distributed process spread across depth.

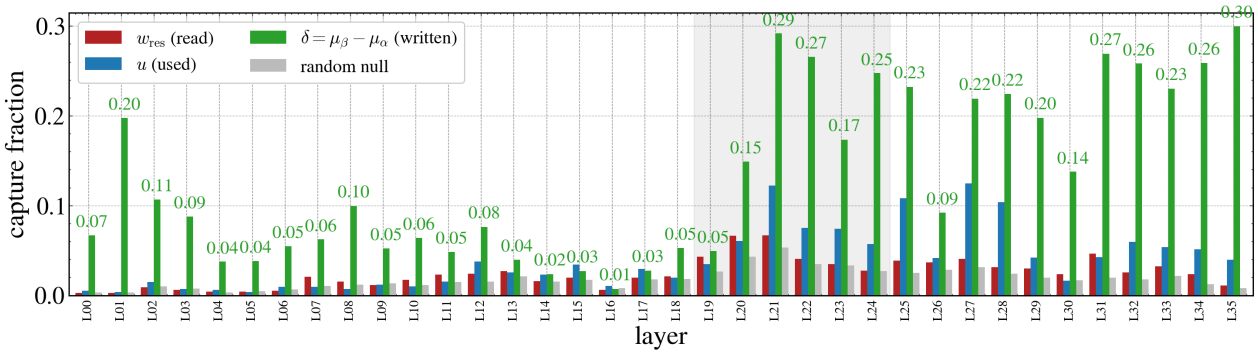


Figure 8: Active feature span capture across layers.

3.6 Clustering

3.6.1 Cluster features and physical meaning

Features are grouped into clusters by their joint causal importance (Section 2.10), and each cluster is enriched for a specific physical feature of the decay. Enrichment is measured by the lift, the ratio of the frequency of a cue feature among the prompts on which the cluster has the largest causal effect to its baseline frequency,

$$\text{lift}(C) = \frac{p(\text{cue} \mid \text{prompts where } C \text{ has the largest ablation effect})}{p(\text{cue})}. \tag{20}$$

A lift of 1 means no enrichment, the cue appearing among the cluster's most affected prompts no more often than at baseline, and a lift well above 1 means strong enrichment. All clusters are enriched for a physical feature with a lift of 6.3 to 18.8 (Table 7), and the features are physically meaningful and ordered by depth, with charge and lepton number in the early layers, nuclear composition and the type of transition in the middle, and the output decision in the late layers.

To determine whether the cluster features detect a physical feature or a class label, we measure their mean activation separately on the prompts of each class. For feature j and class c, the mean activation is

$$\text{fire}_j^{(c)} = \mathbb{E}_{x \in D_c}\left[f_j(x)\right], \tag{21}$$

the average activation of the feature over all prompts of class c, counting prompts on which it is inactive as zero. The class offset of a feature is the difference $\text{fire_diff}_j = \text{fire}_j^{(c_1)} - \text{fire}_j^{(c_0)}$, and Table 7 reports its mean absolute value over the features of each cluster. This activation offset is distinct from a frequency-based specificity, which compares firing frequencies rather than activation magnitudes. A feature that detects a class label activates strongly on one class and weakly on the other, so its class offset is large. A feature that detects a physical feature present in both classes activates comparably on both, so its offset is small relative to the activation level.

The cluster features activate almost equally on the two classes (Table 7). The class offset is small relative to the activation level in every cluster, with the ratio of the mean absolute offset to the mean activation between 0.01 and 0.11. Charge features activate in both α and β contexts because charge is present in both types of decay. They detect charge, not class.

The role labels that the attribution graph assigns to features, such as α-attribution or β-attribution, reflect a contrast of activation weighted by the gradient rather than class-selective activation. A cluster can carry such an attribution role label while its features activate evenly across the two classes, which is exactly what the activation offset of Equation 21 measures directly.

Each cluster is scored on six pre-declared criteria. Directional consistency of its causal effects across prompts, coherence of its decoder directions, semantic specificity of its cue enrichment against a size-matched null, reuse across the surface forms, a joint ablation effect exceeding a size-matched random-feature null, and coverage of more than one difficulty level. A cluster passing

cluster	physical feature	lift	fire α	fire β	mean |offset|	layer	status
1	charge	11.8	2.67	2.77	0.12	L10	accepted
2	lepton family	11.8	2.40	2.35	0.10	L11	partial
3	Z+1 and antineutrino	17.6	1.82	1.83	0.01	L12	descriptive
4	charge -1	7.8	1.97	1.89	0.08	L13	accepted
5	n$\rightarrow$p transition (hub)	18.8	1.94	2.03	0.20	L14–22	partial
6	composite nuclear object	11.8	3.07	3.37	0.29	L15	descriptive
7	charge -1	9.8	0.49	0.46	0.04	L18	descriptive
8	composite nuclear object	17.6	1.36	1.31	0.12	L19	accepted
9	Z+1 and antineutrino	11.8	2.17	1.94	0.23	L20	partial
10	heavy nuclear fragment	15.7	3.06	2.92	0.14	L21	accepted
11	composite nuclear object	11.8	7.75	7.80	0.05	L23	accepted
12	Z+1 output decision	6.3	5.27	5.09	0.39	L24–25	accepted

Table 7: Clusters of the α/β task (HAC $K = 12$), ordered by depth.

all six is accepted, one failing only the causal criterion is partial, and a single-feature cluster is descriptive, which is the status column of Table 7. Of the twelve clusters, six are accepted, three partial, and three descriptive, and the descriptive clusters carry no causal claim in the text. The full validation protocol, the selection of the number of clusters, the per-criterion thresholds, and the cluster ablations are given in the Supplementary Material, Section S.D.

3.6.2 Sequence of physical features

Combining the two facts, that each cluster rotates the writing direction on its own layer and detects a physical feature, the formation of the writing direction can be read as a sequence in depth (Figure 9). The physical features of the decay are inscribed in order. The conserved quantities enter in the early layers, charge at L10 and lepton number at L11. The nuclear composition and the type of transition enter in the middle layers, the neutron-to-proton transition at the L14 to L22 hub and the composition of the nuclear fragment across L15 to L23. The output decision is inscribed in the late layers at L24 to L25. The writing direction does not carry the label α or β. The features write a physical description of the process into the residual stream, and the class label is assembled from it in the later layers.

This gives a mechanistic reading of the rotation of the writing direction with depth (Section 3.2) that complements the feature-level reading of Section 3.5. The writing direction rotates across layers because different clusters are active at different layers and inscribe different physical features. The rotation is therefore not the accumulation of a class signal but a sequence of physical features being written, and the conceptual α or β decision is a late assembly from this physical description.

3.6.3 Structural detection

If the early clusters detect physical features rather than classes, then asking which class the charge cluster writes toward is ill-posed at the level of the representation. Charge is neither α nor β, it is a feature present in both types of decay, and the cluster activates evenly on both. The absence of a pure class signal in the early writing is therefore not a defect of the measurement but a consequence of the early writing encoding the physics of the process rather than the class label.

This is consistent with two earlier results. The use direction, which carries the class decision, is separated from the early writing and assembled near the output (Sections 3.1.3 and 3.5). The attribution role of a feature does not predict the direction of its causal effect (Section 3.6, Supplementary Material, Section S.D). Both reflect the same fact. The class label is not present directly in the early layers. It is computed from the physical features and resolved near the output.

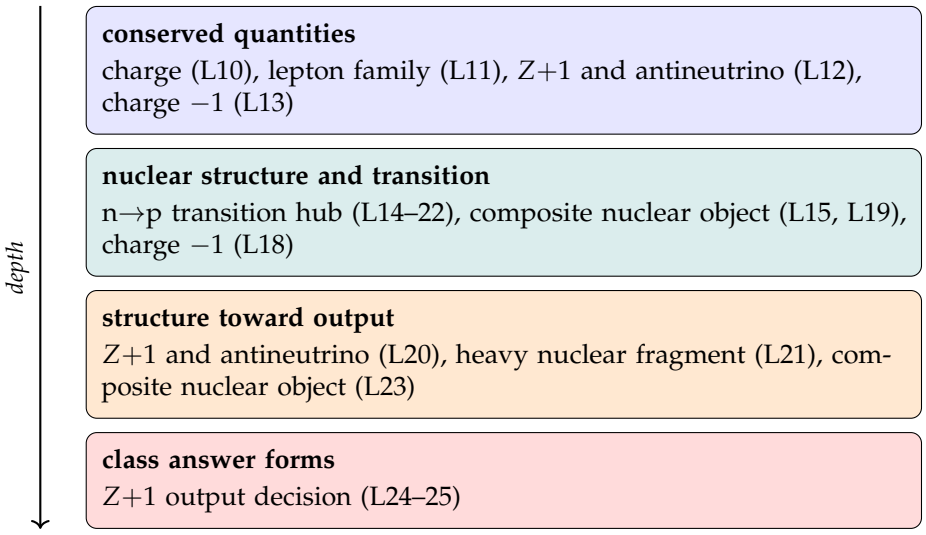


Figure 9: The $\alpha\,/\,\beta$ decay-type circuit organised by depth (HAC).

4 Discussion & Conclusion

4.1 Discussion

Three geometrical objects

The results separate the single notion of a concept direction into three causally and mechanically distinct objects. The reading direction w_{LDA} decodes the concept almost perfectly, with an AUC of 0.98 to 0.99, yet on the α/β task intervention along it leaves the answer unchanged. The inertness is geometric rather than universal. On the particle pairs, where the reading direction has a larger component along the use direction, the same intervention flips the answer, so causal relevance tracks alignment with u rather than decodability. The use direction is causal on every concept tested, and with depth it aligns with the unembedding contrast γ, through which the concept reaches the prediction.

Symmetric dissociation and its mechanism

The dissociation is symmetric. The reading direction carries the class information without causal effect, and the projection onto the use direction decodes barely above chance in the decision zone at the layers where intervention along it reverses the margin on most examples. The within-class dispersion along u links the two halves. It masks the between-class separation along u from a linear reader, and by the attenuation identity of Appendix A it also attenuates the optimal probe along u, so the probe avoids the direction to which the output is sensitive for the same reason that the projection onto that direction decodes poorly.

Qualitative signatures under intervention

Under intervention the writing and use directions dissociate qualitatively. Steering along the writing direction moves the continuation toward a coherent description of the target concept, whereas steering along the use direction raises the target token while the continuation degenerates into repetition. The two directions are also opposite relative to the stream, the writing direction growing in norm on the scaffolded framings while the use direction shrinks on all of them. The specificity of the shift, toward the paired class rather than an arbitrary one, is consistent with the categorical geometry of concepts [38].

Physics behind

The writing direction is formed by the MLP blocks and lies in the span of the active transcoder features well above a random null, so the mechanism that builds it and the dictionary that expresses it describe one process. The features detect the physical properties of the decay, the charge, the lepton number, the nuclear composition, and the transition type, rather than the class, activating comparably on both classes. The model builds a physical description of the process and resolves the class in the later layers, which places the task within the view that an abstract category is computed from more elementary features rather than represented directly [22].

Reservations

Three reservations bound these claims. The use direction is a first-order object, and although the reported reversals are realised through the full nonlinear network, its interpretation beyond the local regime is not guaranteed. The span capture is computed with free coefficients and establishes expressibility by the active dictionary rather than a realised write. All results concern one model and two concept families, so the generality of the trichotomy remains an empirical question.

4.2 Conclusion

The central question of this work was what a concept direction is. The results show that there is no single such object. On a controlled binary task the notion separates into a reading direction, a writing direction, and a use direction, measured independently at every layer and pairwise distinct in geometry, causal effect, and relation to the output. High decodability belongs to the first, the content of the concept to the second, and the causal effect on the answer to the third, and the near orthogonality of reading and use follows from the statistics of the task through the attenuation identity, so a probe fitted to read the class is expected to miss the direction through which the answer is formed.

The transcoder features identify the content of the writing direction. What the model writes is a distributed combination of physical properties of the process, from which the class is assembled in the later layers. For interpretability the consequence is twofold. Decodability, whether of a direction or of a feature, must be paired with a causal test before it is identified with a mechanism, and the content a model writes need not be the category it predicts but the more elementary properties from which that category is computed. Further work includes testing the trichotomy on other model families, replacing the linear reader by nonlinear probes, measuring the realised activation-weighted write of the features, manipulating the task regime to test when the stored content is recruited, and extending the tasks beyond binary classification.

A Whitening

The probe is defined as $w \propto \Sigma^{-1}\Delta\mu$ with the shrunken within-class covariance of Section 2.5.2, and every quantity below is computed on the train split with the same regularisation. For a unit direction $\hat{u}$ write $\sigma_u^2 = \hat{u}^\top \Sigma \hat{u}$. When $\hat{u}$ is an eigenvector of Σ,

$$\cos(w, \hat{u}) = \frac{\langle \Sigma^{-1}\Delta\mu, \hat{u}\rangle}{\|\Sigma^{-1}\Delta\mu\|} = \frac{\langle \Delta\mu, \Sigma^{-1}\hat{u}\rangle}{\|\Sigma^{-1}\Delta\mu\|} = \frac{\langle \Delta\mu, \hat{u}\rangle}{\sigma_u^2\,\|\Sigma^{-1}\Delta\mu\|} = \cos(\Delta\mu, \hat{u})\,\frac{\lambda_{\text{eff}}}{\sigma_u^2}, \qquad \lambda_{\text{eff}} = \frac{\|\Delta\mu\|}{\|\Sigma^{-1}\Delta\mu\|}, \tag{22}$$

using the symmetry of Σ and $\Sigma^{-1}\hat{u} = \hat{u}/\sigma_u^2$, so the alignment of the probe with an axis is the raw alignment of $\Delta\mu$ divided by the dispersion along it. For a general unit $\hat{u}$, with $r = \Sigma\hat{u} - \sigma_u^2\hat{u}$,

$$\cos(w, \hat{u}) = \cos(\Delta\mu, \hat{u})\,\frac{\lambda_{\text{eff}}}{\sigma_u^2} \;-\; \frac{\langle \hat{w}, r\rangle}{\sigma_u^2} \tag{23}$$

holds exactly, the second term being a computable remainder. In the eigenbasis $\|\Sigma^{-1}\Delta\mu\|^2 = \sum_i d_i^2/\lambda_i^2$, so λ_{eff} is a power mean of order minus two, dominated by the smallest eigenvalues $\Delta\mu$ touches, and the ratio $\lambda_{\text{eff}}/\sigma_u^2$ is small for any high-dispersion axis.

The use direction is such an axis. Its dispersion lies at the 98th percentile of the spectrum on both concepts, the attenuation factor is 0.059 and 0.031 in the decision zone, and Equation 22 predicts the measured $\cos(w, u)$ with a per-layer error of 0.039 and 0.044, equal to the measured remainder of 0.038 and 0.048 in Equation 23. The dispersion along u is large because the within-class fluctuations of the margin lie along it. Linearising the margin around the class mean with a prompt-independent gradient $\bar{g}$ gives

$$\text{Var}_c(\text{margin}) = \bar{g}^\top \Sigma\, \bar{g} = \|\bar{g}\|^2 \sigma_u^2, \tag{24}$$

exact at the readout, where the measured ratio of the two sides is 1.06 and 1.16, and a lower bound in the decision zone, where the gradient varies across prompts and the high dispersion of u stands as the measured percentile. The attenuation weakens monotonically with the shrinkage intensity, and at $s \in \{0.03, 0.1, 0.3, 0.6\}$ the conclusions are unchanged.

The high-dimensional null. Small cosines are generic in high dimension, and transformer representations are anisotropic, occupying a narrow cone rather than filling the space [39, 40], so the orthogonality of Section 3.1.3 is read against a covariance-matched null rather than an isotropic one, random directions drawn with the spectrum of Σ, under which the measured cosine sits at the 18th percentile, and against the reliability of the estimated directions, which bounds the disattenuated cosine by 0.074.

B Reproduction of circuit tracing

Attribution graphs over per-layer transcoder features were constructed on four corpora and validated by feature ablation and activation patching. The graphs are compact chains of 13 to 26 features and 19 to 60 edges whose nodes carry readable object features, and the interventions confirm the chains as ensembles, with propagation consistency reaching 1.000 and block patching reproducing a gate effect at 90%, while no single feature or pathway is individually necessary, with per-path necessity of 10 to 28%. The reproduction used per-layer rather than cross-layer transcoders and first-order attributions, and the reconstruction is unreliable in the low-signal regime, which is why the analyses of the main text measure the true residual stream.

Declaration of Use of Autogeneration Tools

Declaration word count: 137 (excluded from the main word count).

Anthropic's Claude, accessed through the Claude Code command-line interface, was used as a coding and analysis assistant throughout this project. It assisted in three ways. First, in writing and refactoring the numbered Python pipeline scripts, including boilerplate for data loading, plotting, and command-line arguments. Second, in cross-checking the numbers reported here against their source data files, by reading the result files and comparing them with the values stated in the report. Third, in editing and consistency-checking the LaTeX manuscript, including notation, cross-references, and the wording of table and figure captions. No prose was generated wholesale and inserted without revision. All research design, the choice of experiments, the interpretation of the results, and the final wording are the author's own, and every number reported in this document is traceable to a committed result file in the accompanying repository.

Bibliography

[1] Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes, 2016. https://arxiv.org/abs/1610.01644.

[2] Thomas M. Cover. Geometrical and statistical properties of systems of linear inequalities with applications in pattern recognition. *IEEE Transactions on Electronic Computers*, EC-14(3):326–334, 1965.

[3] John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of EMNLP-IJCNLP*, 2019. https://arxiv.org/abs/1909.03368.

[4] Tomas Mikolov, Wen-tau Yih, and Geoffrey Zweig. Linguistic regularities in continuous space word representations. In *Proceedings of NAACL-HLT*, 2013.

[5] Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models, 2023. https://arxiv.org/abs/2311.03658.

[6] Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022. https://arxiv.org/abs/2102.12452.

[7] Ian Tenney, Dipanjan Das, and Ellie Pavlick. BERT rediscovers the classical NLP pipeline. In *Proceedings of ACL*, 2019. https://arxiv.org/abs/1905.05950.

[8] John Hewitt and Christopher D. Manning. A structural probe for finding syntax in word representations. In *Proceedings of NAACL-HLT*, 2019.

[9] Fabio Petroni, Tim Rocktäschel, Sebastian Riedel, et al. Language models as knowledge bases? In *Proceedings of EMNLP-IJCNLP*, 2019. https://arxiv.org/abs/1909.01066.

[10] Abhilasha Ravichander, Yonatan Belinkov, and Eduard Hovy. Probing the probing paradigm: Does probing accuracy entail task relevance? In *Proceedings of EACL*, 2021. https://arxiv.org/abs/2005.00719.

[11] Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. Causal abstractions of neural networks, 2021. https://arxiv.org/abs/2106.02997.

[12] Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. LEACE: Perfect linear concept erasure in closed form, 2023. https://arxiv.org/abs/2306.03819.

[13] Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals, 2021. https://arxiv.org/abs/2006.00995.

[14] Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Activation addition: Steering language models without optimization, 2023. https://arxiv.org/abs/2308.10248.

[15] Nina Panickssery, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition, 2023. https://arxiv.org/abs/2312.06681.

[16] Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model, 2023. https://arxiv.org/abs/2306.03341.

[17] Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets, 2023. https://arxiv.org/abs/2310.06824.

[18] Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction, 2024. https://arxiv.org/abs/2406.11717.

[19] Karen Simonyan, Andrea Vedaldi, and Andrew Zisserman. Deep inside convolutional networks: Visualising image classification models and saliency maps, 2014. https://arxiv.org/abs/1312.6034.

[20] Nelson Elhage, Neel Nanda, Catherine Olsson, et al. A mathematical framework for transformer circuits. Transformer Circuits Thread, 2021. https://transformer-circuits.pub/2021/framework/index.html.

[21] Emmanuel Ameisen, Jack Lindsey, Adam Pearce, Wes Gurnee, et al. Circuit tracing: Revealing computational graphs in language models. Transformer Circuits Thread, 2025. https://transformer-circuits.pub/2025/attribution-graphs/methods.html.

[22] Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, et al. On the biology of a large language model. Transformer Circuits Thread, 2025. https://transformer-circuits.pub/2025/attribution-graphs/biology.html.

[23] Jacob Dunefsky, Philippe Chlenski, and Neel Nanda. Transcoders find interpretable LLM feature circuits. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. https://arxiv.org/abs/2406.11944.

[24] Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models, 2023. https://arxiv.org/abs/2309.08600.

[25] Trenton Bricken, Adly Templeton, Joshua Batson, et al. Towards monosemanticity: Decomposing language models with dictionary learning. Transformer Circuits Thread, 2023. https://transformer-circuits.pub/2023/monosemantic-features.

[26] Samuel Marks, Can Rager, Eric J. Michaud, Yonatan Belinkov, David Bau, and Aaron Mueller. Sparse feature circuits: Discovering and editing interpretable causal graphs in language models, 2024. https://arxiv.org/abs/2403.19647.

[27] Nelson Elhage, Tristan Hume, Catherine Olsson, et al. Toy models of superposition. Transformer Circuits Thread, 2022. https://transformer-circuits.pub/2022/toy_model/index.html.

[28] Qwen Team. Qwen3 technical report, 2025. https://arxiv.org/abs/2505.09388.

[29] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. https://arxiv.org/abs/1706.03762.

[30] Michael Hanna. Per-layer transcoders for Qwen3-4B. Hugging Face model repository, 2025. https://huggingface.co/mwhanna/qwen3-4b-transcoders.

[31] Senthooran Rajamanoharan, Tom Lieberum, Nicolas Sonnerat, Arthur Conmy, Vikrant Varma, János Kramár, and Neel Nanda. Jumping ahead: Improving reconstruction fidelity with JumpReLU sparse autoencoders, 2024. https://arxiv.org/abs/2407.14435.

[32] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, et al. Language models are few-shot learners. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020. https://arxiv.org/abs/2005.14165.

[33] Ronald A. Fisher. The use of multiple measurements in taxonomic problems. *Annals of Eugenics*, 7(2):179–188, 1936.

[34] Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, New York, 2006. ISBN 978-0-387-31073-2.

[35] Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.

[36] Jerome H. Friedman. Regularized discriminant analysis. *Journal of the American Statistical Association*, 84(405):165–175, 1989.

[37] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, New York, 2 edition, 2009. ISBN 978-0-387-84857-0.

[38] Kiho Park, Yo Joong Choe, Yibo Jiang, and Victor Veitch. The geometry of categorical and hierarchical concepts in large language models, 2024. https://arxiv.org/abs/2406.01506.

[39] Kawin Ethayarajh. How contextual are contextualized word representations? comparing the geometry of BERT, ELMo, and GPT-2 embeddings. In *Proceedings of EMNLP-IJCNLP*, 2019. https://arxiv.org/abs/1909.00512.

[40] William Timkey and Marten van Schijndel. All bark and no bite: Rogue dimensions in transformer language models obscure representational quality. In *Proceedings of EMNLP*, 2021. https://arxiv.org/abs/2109.04404.